

Statement of Work for Raman Instrument

Background:

The Raman instrument system will have one main function - to analyze samples inside the ICEE (In situ Carbon Evolution Experiments) facility UHV (ultra-high vacuum chamber), see <https://www.nasa.gov/in-situ-carbon-evolution-experiments-icee/> . The Raman instrument will be utilized to analyze samples at various temperatures (300 K to 15 K) and pressures (room pressure to $\sim 10^{-9}$ torr), both in the pristine state and after irradiation via various sources (electrons, UV, X-ray). The instrument will have video/imaging capabilities.

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The Raman spectrometer will be utilized in the preliminary investigation into the chemical evolution of organic compounds and biosignatures bare and covered or mixed in with various ices (H_2O , CH_3OH , etc.), which mimic astrophysical and planetary conditions such as in the ~ 3 -100 AU region of a proto-solar nebula, the surface of Europa, comets or asteroids.

Raman spectroscopy will be used to study the effects of irradiation (UV, electron, X-ray) on condensed organics and biosignatures in planetary ices, measuring destruction rates and determining the formation of products. This information will be incorporated into proto-planetary disk models, planetary mission concept development, biosignature detection and made publicly available via Ramdb (astrochem.org/Ramdb) Raman spectral database.

The primary goal of this work is to determine is to support planetary science work involving the degradation, destruction, and modification of organics and potential biosignatures under astrophysical and planetary science relevant conditions.

The proposed experiments also have a legacy value, as the spectra generated by this proposal will serve as a database (Ramdb) to interpret mission data for the *SHERLOC* instrument on the Mars 2020 rover and the future *Europa* lander mission.

Scope and Specifications:

The Raman instrument system requires the following:

1. One Raman microscope spectrometer system equipped a confocal microscope system for high-end spectroscopic applications.
2. Instrument must have a ZEISS Microscope Body with a 6-fold objective lens revolver
3. Manual or automatic X-Y, and, if necessary, Z, stage system to permit focusing of the Raman instrument inside the ICEE chamber.
4. LED white-light source for Kohler illumination and included external intensity control.
5. Extensible Reflector module changer accepting filter cubes for darkfield, DIC or fluorescence contrast.
6. Camera coupler for sample visualization and including digital video camera, which provides connectivity for white light or fluorescence cameras. The system will include a highly sensitive color video camera.
7. The Raman instrument will permit the simultaneous observation for **sample measurement point and laser spot inside the ICEE UHV chamber**, ensuring the Raman spectrum collected is coming from the intended sample area inside the ICEE UHV.
8. Multi-wavelength laser coupler for up to three excitation sources, permitting switching between all excitation wavelengths (UV to NIR), including automatic alignment of all necessary optical components without manual intervention in the optical beam path.

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9. Computer system for control and data acquisition. Vendor is responsible for ensuring the computer system is acceptable to NASA requirements.
10. 27" or larger, height adjustable Monitor for the computer system.
11. Instrumentation software capable of instrument and experimental control, automated multi-spot measurement sequences with the ability to execute complex scripts at every point, multi-user management for individual measurement pre-configurations, data evaluation and processing components, spectrum peak finder and labeling, and spectra export to Raman database for convenient identification of sample components. The installed spectral data processing software will be capable of spectral subtraction, arithmetic calculations, zoom/trace selection modes, stack display, normalizing, baseline correction (automatic or manual), derivatives, peak detection, peak height (ratio), peak area (ratio), smoothing, spectrum conversion, data removal, wavenumber/wavelength conversion, curve fitting and deconvolution, as well as export of data in publicly accessible, and text file formats.
12. **Instrument will include customized adjustments, as needed, to interface with NASA's ICEE chamber.**
13. Instrument must come with at least one excitation laser source, although multiple are acceptable. If the instrument comes with only one excitation laser source, it must be easily upgradable for two additional excitation laser sources (i.e., total of three excitation sources possible).
14. If the instrument comes with a single excitation laser source, it must be a 532 nm, 30 mW laser source.
15. System must come with a 20x objective lens with super long working distance of 25 mm and a numerical aperture of 0.25.
16. System must come with a 50x objective lens with super long working distance of 18 mm.
17. System must come with an angular objective mount, permitting the microscope to interface with NASA's ICEE chamber.
18. The Raman system will permit real-time display of the spectrum and sample image based on a selected wavenumber intensity.
19. The Raman instrument should come with gratings of 300 and 1800 lines/mm blazed for 500 nm, and capable of spectral resolution (not pixel resolution) at least 1 to 3 cm^{-1} in the 200 to 4000 cm^{-1} range or larger spectral range. This resolution is necessary for the planned research activities.
20. The Raman instrument should come with a thermoelectrically cooled CCD camera with Peltier cooling down to -55 C Chip with 1650 x 200 pixel format.
21. Raman instrument shall come with installation and on-site training as well as free remote online and telephone support by our application scientists.
22. Shipping, packing and handling shall be included in the price.

Period of Performance:

The period of performance shall be 2 years.

NASA POC:

The NASA POC for this purchase is Dr. Andrew Mattioda, phone 650-604-1075; e-mail andrew.mattioda@nasa.gov

Potential Vendor POC:

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A potential vendor POC is Ron Rowe, OXFORD Instruments, phone 978-831-9911, email: Ron.Rowe@oxinst.com, website: <https://raman.oxinst.com/>

Market Research:

Five commercial vendors were contacted regarding this purchase. The vendors include ThermoFisher Scientific, Horiba, Renishaw, Oxford Instruments WiTec, and JASCO. Detailed discussions were had with each vendor representative, identifying the requirements of the Raman instrument. Of these five vendors, the vendors with Raman instruments capable of collecting video images per our chamber requirements were Horiba, Renishaw and Oxford Instruments WiTec. Of these three instruments, only Oxford Instruments/WiTec Raman system was capable of accommodating the conflate port size and vacuum distance requirements to collect both Raman spectra and video images inside NASA's ICEE Ultra-High vacuum chamber.